Optimization 1
ISE 5406

Lecture 3: Real Analysis & Separation Theorems

Dr. Robert Hildebrand

Grado Department of Industrial & Systems Engineering
Virginia Tech

Where We Are in the Course

Last Time (Lecture 2): Introduction to Convexity

- Linear algebra review: combinations, independence, span
- Convex sets: definitions, examples, operations preserving convexity
- Convex and concave functions
- Convex hulls and Carathéodory's theorem
- Extreme points and recession directions

Today (Lecture 3): Real Analysis and Separation

- Closed, open, and compact sets
- Operations preserving closedness and openness
- Existence of minimum/maximum: Weierstrass' theorem
- Separating and supporting hyperplane theorems

Looking Ahead

Next Time (Lecture 4): Convex Functions

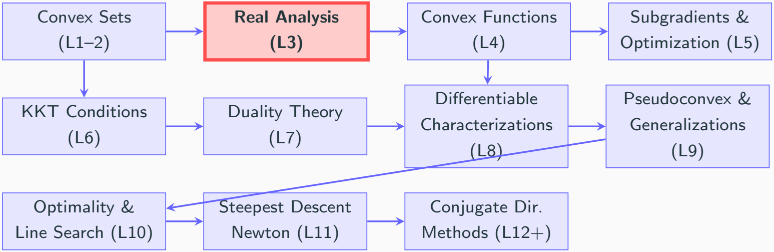
- Strictly convex and strictly concave functions
- Operations preserving convexity of functions
- Continuity of convex functions
- Epigraphs and hypographs

Coming Soon (Lecture 5): Optimality Conditions

- First-order optimality conditions
- Gradients and subgradients
- KKT conditions introduction

Big Picture: Lecture 3 provides the analytical foundations (closed sets, compactness, separation) needed for proving existence of optima and developing optimality conditions.

Big Picture: Course Roadmap



Why Real Analysis and Separation Matter

- Compactness: Guarantees existence of optimal solutions (Weierstrass)
- Separation theorems: Foundation for duality and KKT conditions
- Closedness: Essential for proving convergence of algorithms

Reading for this lecture

- Chapter 2 of Bazaraa, Sherali, and Shetty (2006)
- Chapter 2 and Appendix A of Boyd and Vandenberghe

Outline

Roadmap

Real Analysis

- Closed, Open, and Compact Sets
- Operations Preserving Closedness and Openness of Sets

Existence of Minimum and Maximum

- Infimum and Supremum
- Weierstrass' Theorem

Separating Hyperplanes

- Closest-Point Theorem
- Separating Hyperplane Theorem

Supporting Hyperplanes

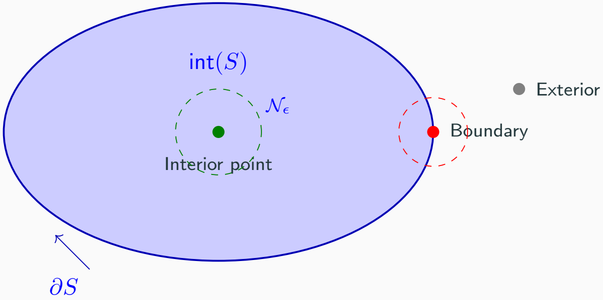
- Supporting Hyperplane Theorem

Definitions: Real Analysis

Let S be an arbitrary set in $\mathbb{R}^n$. (Most definitions extend beyond $\mathbb{R}^n$.)

- **Closure:** $\mathbf{x} \in \mathbb{R}^n$ is said to be in the closure of S, denoted cl(S), if $S \cap \mathcal{N}_\epsilon(\mathbf{x}) \neq \emptyset$ for *every* $\epsilon > 0$.
- **Interior:** $\mathbf{x} \in \mathbb{R}^n$ is said to be in the interior of S, denoted int(S), if $\mathcal{N}_\epsilon(\mathbf{x}) \subset S$ for *some* $\epsilon > 0$. A solid set S is one that has int$(S) \neq \emptyset$.
- **Closed Set:** If $S =$ cl(S), then S is called closed.
- **Open Set:** If $S =$ int(S), then S is called open.
- **Bounded Set:** S is said to be bounded if it can be contained in a ball of a sufficiently large radius centered at the origin.
- **Compact Set:** S is compact if it is *both* closed and bounded.
- **Boundary:** $\mathbf{x} \in \mathbb{R}^n$ is said to be on the boundary of S, denoted ∂S, if for *every* $\epsilon > 0$, $\mathcal{N}_\epsilon(\mathbf{x})$ contains *at least* one point in S and *at least* one point *not* in S.

Visual Summary: Closure, Interior, and Boundary



- Interior $\text{int}(S)$: All points with a neighborhood entirely inside S
- Boundary ∂S: Points where every neighborhood touches both S and S^c
- Closure $\text{cl}(S) = \text{int}(S) \cup \partial S$

Lemma (Closed Sets)
Suppose $S \subseteq \mathbb{R}^n$ is a closed set. Then

1. S contains all its boundary points, i.e., $\partial S \subseteq S$.
2. Its complement $\mathbb{R}^n \backslash S$ is open.
3. For any convergent sequence $\{\mathbf{x}_k\}$ in S with $\lim_k \mathbf{x}_k = \bar{\mathbf{x}}$, we have $\bar{\mathbf{x}} \in S$.

Lemma (Open Sets)
A set $S \subseteq \mathbb{R}^n$ is open if and only if it does not contain any of its boundary points, i.e., $\partial S \cap S = \emptyset$.

Lemma (Boundary)
The boundary of a set $S \subseteq \mathbb{R}^n$ is given by $\partial S = \mathsf{cl}(S) \backslash \mathsf{int}(S)$.

Exercise: Classifying Sets

Classification Challenge

For each set below, determine: (1) Is it open? (2) Is it closed? (3) What is its interior? (4) Is it compact?

Set	**Open/Closed/Compact?**
$\mathbb{R}^n$ and $\emptyset$	
$\{\mathbf{x} \in \mathbb{R}^n : \sum_{i=1}^n x_i^2 \leq 1\}$ (closed ball)	
$\{\mathbf{x} \in \mathbb{R}^n : \sum_{i=1}^n x_i^2 < 1\}$ (open ball)	
$\{\mathbf{x} \in \mathbb{R}^n : \sum_{i=1}^n x_i^2 = 1\}$ (sphere)	
Hyperplane $\{\mathbf{x} : \mathbf{a}^T\mathbf{x} = b\}$	
Halfspace $\{\mathbf{x} : \mathbf{a}^T\mathbf{x} \leq b\}$	

Solution: Classifying Sets

Answers

Set	Classification
$\mathbb{R}^n$ and $\emptyset$	Both open AND closed (only sets with this property!)
Closed ball $\{\mathbf{x} : \|\mathbf{x}\|^2 \leq 1\}$	Closed, not open. Interior = open ball. Compact.
Open ball $\{\mathbf{x} : \|\mathbf{x}\|^2 < 1\}$	Open, not closed. Not compact (not closed).
Sphere $\{\mathbf{x} : \|\mathbf{x}\|^2 = 1\}$	Closed, not open. int$(S) = \emptyset$. Compact.
Hyperplane $\{\mathbf{x} : \mathbf{a}^T\mathbf{x} = b\}$	Closed, not open. int$(S) = \emptyset$. Not compact (unbounded).
Halfspace $\{\mathbf{x} : \mathbf{a}^T\mathbf{x} \leq b\}$	Closed, not open. Not compact (unbounded).

Operations Preserving Closedness and Openness of Sets

Lemma

1. The intersection of any (even infinite) number of closed sets is closed.
2. The intersection of *finitely* many open sets is open.
3. The union of *finitely* many closed sets is closed.
4. The union of any (even infinite) number of open sets is open.

Can you think of *counterexamples* to the following assertions?

1. The intersection of infinitely many open sets is open.
2. The union of infinitely many closed sets is closed.

Supplemental Reference: Walter Rudin's "Principles of Mathematical Analysis"

Exercise: Counterexamples

Think-Pair-Share

Find counterexamples to show that:

1. The intersection of infinitely many **open** sets may not be open.

2. The union of infinitely many **closed** sets may not be closed.

Hint: Consider sets of the form $(-1/n, 1/n)$ *or* $[1/n, 1 - 1/n]$ *for* $n = 1, 2, 3, \ldots$

Solution: Counterexamples

Answers

1. Intersection of open sets may not be open:

   Consider $S_n = (-1/n, 1/n)$ for $n = 1, 2, 3, \ldots$ (all open intervals).

   $$\bigcap_{n=1}^{\infty} S_n = \{0\}$$

   The intersection is a single point, which is closed, not open.

2. Union of closed sets may not be closed:

   Consider $S_n = [1/n, 1 - 1/n]$ for $n = 2, 3, 4, \ldots$ (all closed intervals).

   $$\bigcup_{n=2}^{\infty} S_n = (0, 1)$$

   The union is an open interval, which is open, not closed.

Convexity and Closure/Interior

Theorem
Let S be a convex set in $\mathbb{R}^n$ with a nonempty interior. Let $\mathbf{x}_1 \in \mathrm{cl}(S)$ and $\mathbf{x}_2 \in \mathrm{int}(S)$. Then

$$\lambda \mathbf{x}_1 + (1-\lambda)\mathbf{x}_2 \in \mathrm{int}(S) \text{ for } \textit{each } \lambda \in (0,1).$$

Proof: reading assignment (see Theorem 2.2.2 of Bazaraa et al.)

Corollaries

1. Let S be a convex set. Then $\mathrm{int}(S)$ is convex.
2. Let S be a convex set with $\mathrm{int}(S) \neq \emptyset$. Then $\mathrm{cl}(S)$ is convex.
3. Let S be a convex set with $\mathrm{int}(S) \neq \emptyset$. Then $\mathrm{cl}(\mathrm{int}(S)) = \mathrm{cl}(S)$.
4. Let S be a convex set with $\mathrm{int}(S) \neq \emptyset$. Then $\mathrm{int}(\mathrm{cl}(S)) = \mathrm{int}(S)$.

We can replace $\mathrm{int}(S)$ with the relative interior of S in the above results.

Roadmap

Infimum and Supremum

Let S be an arbitrary set in $\mathbb{R}$.

Infimum: A scalar $l \in \mathbb{R}$ is called the infimum or the greatest lower bound of S, denoted $\inf(S)$, if

1. l is a lower bound of S, i.e., $l \leq x$ for each $x \in S$,
2. for all lower bounds b of S, $l \geq b$.

Convention: $\inf(\emptyset) = +\infty$, $\inf(S) = -\infty$ if S is unbounded from below

Supremum: A scalar $u \in \mathbb{R}$ is called the supremum or the least upper bound of S, denoted $\sup(S)$, if

1. u is an upper bound of S, i.e., $u \geq x$ for each $x \in S$,
2. for all upper bounds b of S, $u \leq b$.

Convention: $\sup(\emptyset) = -\infty$, $\sup(S) = +\infty$ if S is unbounded from above

Exercise: Infimum vs Minimum

Quick Check

For each set, find the infimum and determine if the minimum exists.

1. $S = (0, 1]$
2. $S = [0, 1]$
3. $S = \{1/n : n \in \mathbb{N}\}$
4. $S = \{x \in \mathbb{R} : x^2 > 2\}$

Solution: Infimum vs Minimum

Answers

1. $S = (0, 1]$: $\inf(S) = 0$, but minimum does not exist (0 is not in S).
2. $S = [0, 1]$: $\inf(S) = 0$, and $\min(S) = 0$ exists (0 is in S).
3. $S = \{1/n : n \in \mathbb{N}\} = \{1, 1/2, 1/3, \dots\}$:

   $\inf(S) = 0$, but minimum does not exist (0 is not achieved by any $1/n$).
4. $S = \{x \in \mathbb{R} : x^2 > 2\} = (-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$:

   $\inf(S) = -\infty$ (unbounded below), so minimum does not exist.

Key insight: The infimum always exists (in the extended reals), but the minimum exists only if the infimum is attained by some element of S.

Existence of Minimum and Maximum

Let $S \subseteq \mathbb{R}^n$ be a nonempty set and $f : S \to \mathbb{R}$.

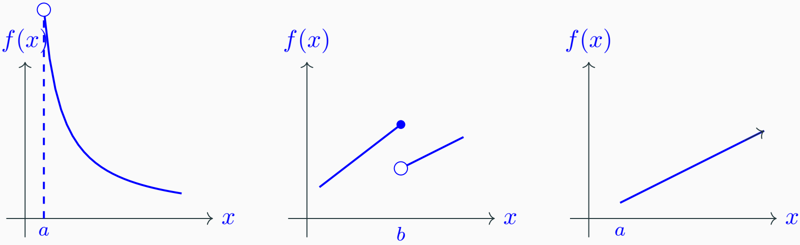
When do the problems

$$\inf_{\mathbf{x}\in S} f(\mathbf{x}) \Big/ \min_{\mathbf{x}\in S} f(\mathbf{x}) \qquad \text{and} \qquad \sup_{\mathbf{x}\in S} f(\mathbf{x}) \Big/ \max_{\mathbf{x}\in S} f(\mathbf{x})$$

admit (optimal) solutions?

- $\bar{\mathbf{x}}$ is a minimizing solution for the problem $\inf\{f(\mathbf{x}) : \mathbf{x} \in S\}$, provided that $\bar{\mathbf{x}} \in S$ and $f(\bar{\mathbf{x}}) \leq f(\mathbf{x})$ for all $\mathbf{x} \in S$.
  If a minimizing solution exists, we simply write $\min_{\mathbf{x}\in S} f(\mathbf{x})$.
- $\bar{\mathbf{x}}$ is a maximizing solution for the problem $\sup\{f(\mathbf{x}) : \mathbf{x} \in S\}$, provided that $\bar{\mathbf{x}} \in S$ and $f(\bar{\mathbf{x}}) \geq f(\mathbf{x})$ for all $\mathbf{x} \in S$.
  If a maximizing solution exists, we simply write $\max_{\mathbf{x}\in S} f(\mathbf{x})$.

Why Optima May Not Exist



(a) S not closed (b) f discontinuous (c) S unbounded

(a) S is not closed: infimum not attained at boundary

(b) f is not continuous: jump at discontinuity

(c) S is unbounded: function keeps decreasing (or increasing)

Weierstrass' Theorem

Weierstrass' Theorem (Sufficient Conditions for Existence of Min./Max.)

Let S be a nonempty, compact (closed and bounded) set, and let $f : S \to \mathbb{R}$ be continuous on S. Then the problem $\min\{f(\mathbf{x}) : \mathbf{x} \in S\}$ attains its minimum.

Key Points:

- Compactness = closed + bounded
- Continuity of f is essential
- The theorem guarantees *existence*, not uniqueness
- Same result holds for maximum

Why This Matters for Optimization:

- Provides conditions under which we *know* an optimal solution exists
- Justifies writing "$\min$" instead of "$\inf$" when conditions are met
- Foundation for convergence analysis of optimization algorithms

Exercise: Applying Weierstrass' Theorem

Application

For each optimization problem, determine if Weierstrass' theorem guarantees a solution exists. If not, explain which condition fails.

1. $\min\{x^2 : x \in [-1, 1]\}$
2. $\min\{1/x : x \in (0, 1]\}$
3. $\min\{e^{-x} : x \in [0, \infty)\}$
4. $\min\{x_1^2 + x_2^2 : x_1^2 + x_2^2 \leq 4\}$

Solution: Applying Weierstrass' Theorem

Answers

1. $\min\{x^2 : x \in [-1,1]\}$

   Yes, solution exists. $S = [-1,1]$ is compact, $f(x) = x^2$ is continuous. $\min = 0$ at $x = 0$.

2. $\min\{1/x : x \in (0,1]\}$

   No guarantee. $S = (0,1]$ is not closed (missing 0). Indeed, $\inf = 1$ but no minimum.

3. $\min\{e^{-x} : x \in [0,\infty)\}$

   No guarantee. $S = [0,\infty)$ is not bounded. Indeed, $\inf = 0$ but no minimum (approaches 0 as $x \to \infty$).

4. $\min\{x_1^2 + x_2^2 : x_1^2 + x_2^2 \leq 4\}$

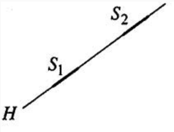
   Yes, solution exists. S is the closed ball of radius 2 (compact), f is continuous. $\min = 0$ at origin.

Roadmap

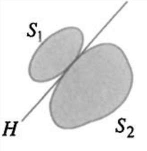
Separating Hyperplane

Let S_1 and S_2 be nonempty sets in $\mathbb{R}^n$.

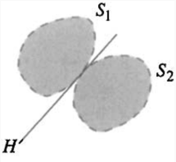
- A hyperplane $H := \{\mathbf{x} \in \mathbb{R}^n : \mathbf{p}^T\mathbf{x} = \alpha\}$ separates S_1 and S_2 if $\mathbf{p}^T\mathbf{x} \geq \alpha$ for each $\mathbf{x} \in S_1$ and $\mathbf{p}^T\mathbf{x} \leq \alpha$ for each $\mathbf{x} \in S_2$.
- If, additionally, $S_1 \cup S_2 \not\subset H$, then H properly separates S_1 and S_2.
- The hyperplane H strictly separates S_1 and S_2 if $\mathbf{p}^T\mathbf{x} > \alpha$ for each $\mathbf{x} \in S_1$ and $\mathbf{p}^T\mathbf{x} < \alpha$ for each $\mathbf{x} \in S_2$.
- The hyperplane H strongly separates S_1 and S_2 if for *some* $\epsilon > 0$, $\mathbf{p}^T\mathbf{x} \geq \alpha + \epsilon$ for each $\mathbf{x} \in S_1$ and $\mathbf{p}^T\mathbf{x} \leq \alpha$ for each $\mathbf{x} \in S_2$.



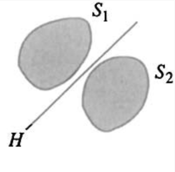
Improper separation



Proper separation



Strict separation



Strong separation

Exercise: Types of Separation

Conceptual Question

Consider the following pairs of sets in $\mathbb{R}^2$. For each pair, determine the strongest type of separation possible (none, improper, proper, strict, or strong).

1. $S_1 = \{(x, y) : x \geq 0\}$ and $S_2 = \{(x, y) : x \leq 0\}$
2. $S_1 = \{(x, y) : x > 0\}$ and $S_2 = \{(x, y) : x < 0\}$
3. $S_1 = \{(x, y) : x \geq 1\}$ and $S_2 = \{(x, y) : x \leq 0\}$

Solution: Types of Separation

Answers

1. $S_1 = \{(x, y) : x \geq 0\}$ and $S_2 = \{(x, y) : x \leq 0\}$

   The hyperplane $x = 0$ separates them, but $S_1 \cup S_2 = \mathbb{R}^2 \supset H$.

   Improper separation (both sets share the boundary).

2. $S_1 = \{(x, y) : x > 0\}$ and $S_2 = \{(x, y) : x < 0\}$

   The hyperplane $x = 0$ separates them with $\mathbf{p}^T\mathbf{x} > 0$ for S_1 and < 0 for S_2.

   Strict separation (but not strong—no gap ϵ).

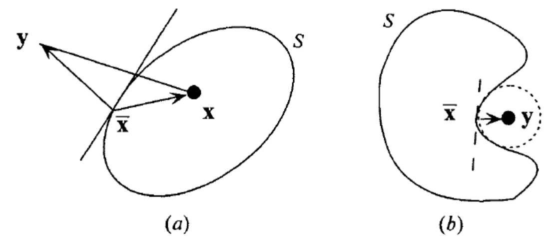
3. $S_1 = \{(x, y) : x \geq 1\}$ and $S_2 = \{(x, y) : x \leq 0\}$

   Take $x = 0.5$. Then $\mathbf{p}^T\mathbf{x} \geq 1 \geq 0.5 + 0.5$ for S_1 and $\leq 0 < 0.5$ for S_2.

   Strong separation (there's a gap of width 1 between the sets).

Closest-Point Theorem
Given a nonempty closed convex set $S \subseteq \mathbb{R}^n$ and a point $\mathbf{y} \in \mathbb{R}^n$ with $\mathbf{y} \notin S$, there exists a unique point $\bar{\mathbf{x}} \in S$ with minimum distance from $\mathbf{y}$.
Moreover, $\bar{\mathbf{x}}$ is the minimizing point if and only if

$$(\mathbf{y} - \bar{\mathbf{x}})^T (\mathbf{x} - \bar{\mathbf{x}}) \leq 0, \text{ for all } \mathbf{x} \in S.$$



Exercise: Closest Point

Computation

Find the closest point in S to the given point $\mathbf{y}$.

1. $S = \{\mathbf{x} \in \mathbb{R}^2 : x_1 + x_2 \leq 1, x_1 \geq 0, x_2 \geq 0\}$, $\mathbf{y} = (2, 2)^T$

2. $S = \{\mathbf{x} \in \mathbb{R}^2 : x_1^2 + x_2^2 \leq 1\}$, $\mathbf{y} = (3, 4)^T$

Hint: For problem 2, the closest point lies on the boundary in the direction of $\mathbf{y}$ *from the origin.*

Solution: Closest Point

Answers

1. $S = \{\mathbf{x} \in \mathbb{R}^2 : x_1 + x_2 \leq 1, x_1 \geq 0, x_2 \geq 0\}$, $\mathbf{y} = (2,2)^T$

   The closest point is on the hypotenuse $x_1 + x_2 = 1$.

   Project $(2,2)$ onto the line $x_1 + x_2 = 1$:

   Direction: $\mathbf{n} = (1,1)/\sqrt{2}$. Distance to line: $(2+2-1)/\sqrt{2} = 3/\sqrt{2}$.

   $\bar{\mathbf{x}} = (2,2) - \frac{3}{\sqrt{2}} \cdot \frac{(1,1)}{\sqrt{2}} = (2,2) - (1.5, 1.5) = (0.5, 0.5)^T$

2. $S = \{\mathbf{x} \in \mathbb{R}^2 : x_1^2 + x_2^2 \leq 1\}$, $\mathbf{y} = (3,4)^T$

   The closest point is on the boundary (unit circle) in the direction of $\mathbf{y}$.

   $\|\mathbf{y}\| = \sqrt{9+16} = 5$

   $\bar{\mathbf{x}} = \frac{\mathbf{y}}{\|\mathbf{y}\|} = (3/5, 4/5)^T = (0.6, 0.8)^T$

Separating Hyperplane Theorem

Theorem (Separation of a Convex Set and a Point)
Let S be a nonempty closed convex set in $\mathbb{R}^n$ and $\mathbf{y} \notin S$. Then, there exists a nonzero vector $\mathbf{p}$ and a scalar α such that

$$\mathbf{p}^T\mathbf{y} > \alpha \text{ and } \mathbf{p}^T\mathbf{x} \leq \alpha \text{ for each } \mathbf{x} \in S.$$

Corollaries

1. Let S be a closed convex set in $\mathbb{R}^n$. Then S is the intersection of all halfspaces containing S.
2. Let S be a nonempty set, and let $\mathbf{y} \notin \text{cl}(\text{conv}(S))$. Then there exists a strongly separating hyperplane separating S and $\{\mathbf{y}\}$.
3. Farkas' Lemma: Let $A \in \mathbb{R}^{m\times n}$ and $c \in \mathbb{R}^n$. Then exactly one of the following two systems has a solution:

   System 1: $A\mathbf{x} \leq \mathbf{0}$ and $\mathbf{c}^T\mathbf{x} > 0$ for some $\mathbf{x} \in \mathbb{R}^n$.

   System 2: $A^T\mathbf{y} = \mathbf{c}$ and $\mathbf{y} \geq \mathbf{0}$ for some $\mathbf{y} \in \mathbb{R}^m$.

Roadmap

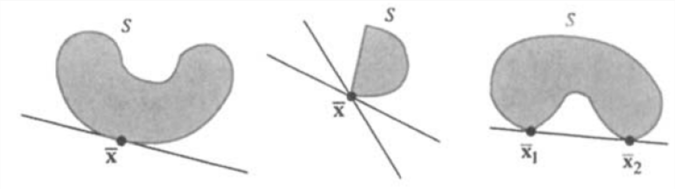
Supporting Hyperplane

Let S be a nonempty set in $\mathbb{R}^n$ and $\bar{\mathbf{x}} \in \partial S$.

- A hyperplane $H := \{\mathbf{x} \in \mathbb{R}^n : \mathbf{p}^T(\mathbf{x} - \bar{\mathbf{x}}) = 0\}$ is called a supporting hyperplane of S at $\bar{\mathbf{x}}$ if S is *entirely* contained in one of the halfspaces

$$\{\mathbf{x} \in \mathbb{R}^n : \mathbf{p}^T(\mathbf{x} - \bar{\mathbf{x}}) \leq 0\}, \quad \text{or} \quad \{\mathbf{x} \in \mathbb{R}^n : \mathbf{p}^T(\mathbf{x} - \bar{\mathbf{x}}) \geq 0\}$$

- If $S \not\subseteq H$, then H is a proper supporting hyperplane of S at $\bar{\mathbf{x}}$.



Supporting Hyperplane Theorem

Theorem (Supporting Hyperplane)
Let S be a nonempty convex set in $\mathbb{R}^n$ and $\bar{\mathbf{x}} \in \partial S$. Then there exists a hyperplane that supports S at $\bar{\mathbf{x}}$. That is, there exists $\mathbf{p} \neq \mathbf{0}$ such that $\mathbf{p}^T(\mathbf{x} - \bar{\mathbf{x}}) \leq 0$ for each $\mathbf{x} \in \text{cl}(S)$.

Geometric Interpretation:

- At every boundary point of a convex set, we can find a hyperplane that "touches" the set without crossing into it
- The set lies entirely on one side of the supporting hyperplane
- This is a key property that characterizes convex sets

Application: Supporting hyperplanes are fundamental to:

- Characterizing optimal solutions (KKT conditions)
- Duality theory
- Cutting plane methods in optimization

Exercise: Supporting Hyperplanes

Geometric Construction

For each set and boundary point, find a supporting hyperplane.

1. $S = \{\mathbf{x} \in \mathbb{R}^2 : x_1^2 + x_2^2 \leq 1\}$ at $\bar{\mathbf{x}} = (1, 0)^T$

2. $S = \{\mathbf{x} \in \mathbb{R}^2 : x_1 \geq 0, x_2 \geq 0, x_1 + x_2 \leq 1\}$ at $\bar{\mathbf{x}} = (0, 0)^T$

3. $S = \{\mathbf{x} \in \mathbb{R}^2 : x_1 \geq 0, x_2 \geq 0, x_1 + x_2 \leq 1\}$ at $\bar{\mathbf{x}} = (0.5, 0.5)^T$

Solution: Supporting Hyperplanes

Answers

1. $S = \{\mathbf{x} : x_1^2 + x_2^2 \leq 1\}$ at $\bar{\mathbf{x}} = (1,0)^T$

   The gradient of $g(\mathbf{x}) = x_1^2 + x_2^2 - 1$ at $(1,0)$ is $(2,0)^T$.

   Supporting hyperplane: $x_1 = 1$ (or $\mathbf{p} = (1,0)^T$).

2. $S = \{\mathbf{x} : x_1 \geq 0, x_2 \geq 0, x_1 + x_2 \leq 1\}$ at $\bar{\mathbf{x}} = (0,0)^T$

   At the corner $(0,0)$, multiple supporting hyperplanes exist!

   Any $\mathbf{p}$ with $p_1 \leq 0$ and $p_2 \leq 0$ works.

   Examples: $x_1 = 0$, $x_2 = 0$, or $x_1 + x_2 = 0$.

3. $S = \{\mathbf{x} : x_1 \geq 0, x_2 \geq 0, x_1 + x_2 \leq 1\}$ at $\bar{\mathbf{x}} = (0.5, 0.5)^T$

   This point is on the edge $x_1 + x_2 = 1$.

   Supporting hyperplane: $x_1 + x_2 = 1$ (or $\mathbf{p} = (1,1)^T$).

End-of-Lecture Assessment

Quick Assessment (5 questions)
Answer True or False:

1. **T/F:** Every bounded set is compact.
2. **T/F:** The interior of a convex set is convex.
3. **T/F:** Weierstrass' theorem guarantees that $\min\{1/x : x \in (0, 1]\}$ attains its minimum.
4. **T/F:** Every pair of disjoint convex sets can be strictly separated.
5. **T/F:** At every boundary point of a convex set, there exists a supporting hyperplane.

End-of-Lecture Assessment: Solutions

Answers

1. False. Compact = closed AND bounded. Example: $(0, 1)$ is bounded but not closed, hence not compact.
2. True. This is Corollary 1 from the theorem about convex sets and interior/closure.
3. False. $(0, 1]$ is not closed (hence not compact), so Weierstrass doesn't apply. Indeed, $\inf = 1$ but the minimum is not attained.
4. False. Consider $S_1 = \{(x, y) : x < 0\}$ and $S_2 = \{(x, y) : x > 0\} \cup \{(0, 0)\}$. They're disjoint but can only be improperly separated.
5. True. This is exactly the Supporting Hyperplane Theorem.

Summary and Next Steps

Key Takeaways

- Closed sets contain all boundary points; open sets contain none
- Compact = closed + bounded (essential for existence of optima)
- Weierstrass' theorem: Continuous function on compact set attains min/max
- Separating hyperplanes: Can separate a point from a closed convex set
- Supporting hyperplanes: Exist at every boundary point of a convex set

Next Time:

- Strictly convex and strictly concave functions
- Operations preserving convexity of functions
- Epigraphs and their connection to convex sets

Questions?